

Second-Order Fluctuations of the Accentuation Alignment Ratio Current Noise-Conditional Correction and the Full Random-Design Problem

Theory supplement for `AccentuationPredRMT`

August 18, 2026

Abstract

The own-path accentuation metrics depend on the nonlinear ratio

$$R = \frac{\widehat{\beta}^\top \beta^*}{\widehat{\beta}^\top \widehat{\beta}}.$$

Leading deterministic equivalents replace this random ratio by a ratio of deterministic first and second moments. The current code adds an approximate second-order correction for response-noise fluctuations. This note derives that correction from a conditional Gaussian calculation and shows exactly where its spectral sums arise. It then formulates the full unconditional variance problem, which must additionally retain random-design resolvent fluctuations and, for cross-validated ridge, randomness of the selected regularization.

1 Ridge estimator and accentuation ratio

Consider

$$\mathbf{x}_i \sim \mathcal{N}(0, \Sigma), \quad y_i = (\beta^*)^\top \mathbf{x}_i + \sigma \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 1). \quad (1)$$

For centered data, let

$$\widehat{\Sigma} = \frac{X^\top X}{n}, \quad Q_X = (\widehat{\Sigma} + \lambda I)^{-1}. \quad (2)$$

The normalized-loss ridge estimator is

$$\widehat{\beta} = Q_X \frac{X^\top \mathbf{y}}{n} \quad (3)$$

$$= Q_X \widehat{\Sigma} \beta^* + \frac{\sigma}{n} Q_X X^\top \boldsymbol{\epsilon}. \quad (4)$$

Define the sample ridge filter

$$P_X = Q_X \widehat{\Sigma} = \widehat{\Sigma} (\widehat{\Sigma} + \lambda I)^{-1}. \quad (5)$$

Conditional on X , Eq. (4) becomes

$$\widehat{\beta} = \boldsymbol{\mu}_X + \boldsymbol{\xi}_X, \quad \boldsymbol{\mu}_X = P_X \beta^*, \quad (6)$$

with

$$\boldsymbol{\xi}_X \mid X \sim \mathcal{N}(0, \Omega_X), \quad \Omega_X = \frac{\sigma^2}{n} Q_X \widehat{\Sigma} Q_X. \quad (7)$$

For one fitted model, the true-on-predicted slope along its own pixel-gradient path is

$$R = \frac{N}{D}, \quad N = \widehat{\boldsymbol{\beta}}^\top \boldsymbol{\beta}^*, \quad D = \widehat{\boldsymbol{\beta}}^\top \widehat{\boldsymbol{\beta}}. \quad (8)$$

The normalized accentuation error and pathwise coefficient of determination are

$$\frac{E_{\text{acc}}}{S} = (1 - R)^2, \quad S = (\boldsymbol{\beta}^*)^\top \Sigma \boldsymbol{\beta}^*, \quad (9)$$

$$R_{\text{acc}}^2 = g(R) = 1 - \left(1 - \frac{1}{R}\right)^2. \quad (10)$$

2 Exact Gaussian moments conditional on the design

Using Eq. (6),

$$N = (\boldsymbol{\beta}^*)^\top \boldsymbol{\mu}_X + (\boldsymbol{\beta}^*)^\top \boldsymbol{\xi}_X, \quad (11)$$

$$D = \boldsymbol{\mu}_X^\top \boldsymbol{\mu}_X + 2\boldsymbol{\mu}_X^\top \boldsymbol{\xi}_X + \boldsymbol{\xi}_X^\top \boldsymbol{\xi}_X. \quad (12)$$

Their conditional means are

$$n_X = \mathbb{E}_\epsilon[N \mid X] = (\boldsymbol{\beta}^*)^\top \boldsymbol{\mu}_X, \quad (13)$$

$$d_X = \mathbb{E}_\epsilon[D \mid X] = \boldsymbol{\mu}_X^\top \boldsymbol{\mu}_X + \text{tr } \Omega_X. \quad (14)$$

The centered fluctuations are

$$\delta N = (\boldsymbol{\beta}^*)^\top \boldsymbol{\xi}_X, \quad (15)$$

$$\delta D = 2\boldsymbol{\mu}_X^\top \boldsymbol{\xi}_X + \left(\boldsymbol{\xi}_X^\top \boldsymbol{\xi}_X - \text{tr } \Omega_X\right). \quad (16)$$

For a centered Gaussian vector $\boldsymbol{\xi} \sim \mathcal{N}(0, \Omega)$,

$$\text{Var}(\boldsymbol{a}^\top \boldsymbol{\xi}) = \boldsymbol{a}^\top \Omega \boldsymbol{a}, \quad (17)$$

$$\text{Var}(\boldsymbol{\xi}^\top \boldsymbol{\xi}) = 2 \text{tr}(\Omega^2), \quad (18)$$

$$\text{Cov}(\boldsymbol{a}^\top \boldsymbol{\xi}, \boldsymbol{\xi}^\top \boldsymbol{\xi}) = 0. \quad (19)$$

The last equality follows because centered third-order Gaussian moments vanish. Consequently,

$$\text{Var}_\epsilon(N \mid X) = (\boldsymbol{\beta}^*)^\top \Omega_X \boldsymbol{\beta}^*, \quad (20)$$

$$\text{Cov}_\epsilon(N, D \mid X) = 2(\boldsymbol{\beta}^*)^\top \Omega_X \boldsymbol{\mu}_X, \quad (21)$$

$$\text{Var}_\epsilon(D \mid X) = 4\boldsymbol{\mu}_X^\top \Omega_X \boldsymbol{\mu}_X + 2 \text{tr}(\Omega_X^2). \quad (22)$$

Equations (20)–(22) are exact conditional Gaussian identities.

3 Delta-method variance at fixed design

For $f(N, D) = N/D$, the gradient at (n_X, d_X) is

$$\nabla f = \begin{pmatrix} d_X^{-1} \\ -n_X d_X^{-2} \end{pmatrix} = \frac{1}{d_X} \begin{pmatrix} 1 \\ -R_X \end{pmatrix}, \quad R_X = \frac{n_X}{d_X}. \quad (23)$$

The first-order delta approximation therefore gives

$$\text{Var}_\epsilon(R \mid X) \simeq \frac{\text{Var}_\epsilon(N \mid X) - 2R_X \text{Cov}_\epsilon(N, D \mid X) + R_X^2 \text{Var}_\epsilon(D \mid X)}{d_X^2} \quad (24)$$

$$= \frac{\mathbf{v}_X^\top \Omega_X \mathbf{v}_X + 2R_X^2 \text{tr}(\Omega_X^2)}{d_X^2}, \quad (25)$$

$$\mathbf{v}_X = \boldsymbol{\beta}^* - 2R_X \boldsymbol{\mu}_X. \quad (26)$$

This compact expression is the source of the current correction. Its first term is the variance of a linear Gaussian form; its second is the variance of the centered quadratic form in the denominator.

4 Deterministic substitutions made by the current code

Let

$$\Sigma = U \text{diag}(s_1, \dots, s_d) U^\top, \quad b_k = \mathbf{u}_k^\top \boldsymbol{\beta}^*. \quad (27)$$

The effective regularization $\kappa = \kappa(\lambda)$ solves

$$\kappa - \lambda = \frac{\kappa}{n} \sum_k \frac{s_k}{s_k + \kappa}. \quad (28)$$

Define

$$t_k = \frac{s_k}{s_k + \kappa}, \quad (29)$$

$$\text{df}_2 = \sum_k \frac{s_k^2}{(s_k + \kappa)^2}, \quad (30)$$

$$C_{\text{sig}} = \sum_k \frac{s_k b_k^2}{(s_k + \kappa)^2}. \quad (31)$$

The first-order deterministic numerator and denominator used by the code are

$$N_{\text{det}} = \sum_k t_k b_k^2, \quad (32)$$

$$D_{\text{det}} = \sum_k t_k^2 b_k^2 + \frac{\kappa^2 C_{\text{sig}} + \sigma^2}{n - \text{df}_2} \sum_k \frac{s_k}{(s_k + \kappa)^2}, \quad (33)$$

$$R_{\text{det}} = \frac{N_{\text{det}}}{D_{\text{det}}}. \quad (34)$$

These equations are implemented by `accentuation.alignment` in `rmt_core/ridge_theory_lib.py`.

For the variance correction, the current implementation makes the simplified substitutions

$$\boldsymbol{\mu}_X \rightsquigarrow T_\kappa \boldsymbol{\beta}^*, \quad T_\kappa = \Sigma(\Sigma + \kappa I)^{-1}, \quad (35)$$

$$\Omega_X \rightsquigarrow \frac{\sigma^2}{n} H_\kappa, \quad H_\kappa = \Sigma(\Sigma + \kappa I)^{-2}, \quad (36)$$

$$(R_X, d_X) \rightsquigarrow (R_{\text{det}}, D_{\text{det}}). \quad (37)$$

This is a hybrid leading-DE plus conditional-noise approximation. It is not a full central-limit theorem for the random design.

4.1 Linear Gaussian contribution

Under Eqs. (35)–(36),

$$\mathbf{v}_{\text{det}} = (I - 2R_{\text{det}}T_{\kappa})\boldsymbol{\beta}^*. \quad (38)$$

Therefore

$$\mathbf{v}_{\text{det}}^{\top} \left(\frac{\sigma^2}{n} H_{\kappa} \right) \mathbf{v}_{\text{det}} = \frac{\sigma^2}{n} \sum_k \frac{s_k b_k^2}{(s_k + \kappa)^2} \left(1 - 2R_{\text{det}} \frac{s_k}{s_k + \kappa} \right)^2 \quad (39)$$

$$= \frac{\sigma^2}{n} (C_{\text{sig}} - 4R_{\text{det}}M_1 + 4R_{\text{det}}^2M_2), \quad (40)$$

where

$$M_1 = \sum_k \frac{s_k^2 b_k^2}{(s_k + \kappa)^3}, \quad (41)$$

$$M_2 = \sum_k \frac{s_k^3 b_k^2}{(s_k + \kappa)^4}. \quad (42)$$

Thus the apparently specialized combination $C_{\text{sig}} - 4R_{\text{det}}M_1 + 4R_{\text{det}}^2M_2$ is simply the eigenbasis expansion of a single quadratic form.

4.2 Quadratic Gaussian contribution

The second term in Eq. (25) becomes

$$2R_{\text{det}}^2 \text{tr} \left[\left(\frac{\sigma^2}{n} H_{\kappa} \right)^2 \right] = \frac{2R_{\text{det}}^2 \sigma^4}{n^2} T_{\text{quad}}, \quad (43)$$

$$T_{\text{quad}} = \sum_k \frac{s_k^2}{(s_k + \kappa)^4}. \quad (44)$$

4.3 Current implemented formula

Combining Eqs. (40) and (44) gives

$$\boxed{\text{Var}_{\text{current}}(R) = \frac{\frac{\sigma^2}{n} (C_{\text{sig}} - 4R_{\text{det}}M_1 + 4R_{\text{det}}^2M_2) + \frac{2R_{\text{det}}^2 \sigma^4}{n^2} T_{\text{quad}}}{D_{\text{det}}^2}}. \quad (45)$$

Equation (45) is implemented by `accentuation_var_R` in `rmt_core/ridge_theory_lib.py`.

5 How the current variance correction is used

The exact bias–variance identity

$$\mathbb{E}[(1 - R)^2] = (1 - \mathbb{E}R)^2 + \text{Var}(R) \quad (46)$$

motivates the code approximation

$$\frac{\mathbb{E}E_{\text{acc}}}{S} \simeq (1 - R_{\text{det}})^2 + \text{Var}_{\text{current}}(R). \quad (47)$$

This is enabled by `include_var_R=True`.

For pathwise R^2 , let

$$g(r) = 1 - \left(1 - \frac{1}{r}\right)^2 = \frac{2}{r} - \frac{1}{r^2}. \quad (48)$$

A second-order scalar delta expansion gives

$$\mathbb{E}[g(R)] \simeq g(R_{\text{det}}) + \frac{1}{2}g''(R_{\text{det}}) \text{Var}_{\text{current}}(R), \quad (49)$$

$$\frac{1}{2}g''(r) = \frac{2r - 3}{r^4}. \quad (50)$$

Hence

$$R_{\text{acc}}^{2,\text{corrected}} = 1 - \left(1 - \frac{1}{R_{\text{det}}}\right)^2 \frac{2R_{\text{det}} - 3}{R_{\text{det}}^4} \text{Var}_{\text{current}}(R). \quad (51)$$

This is enabled by `include_delta_correction=True`. Equation (51) is poorly conditioned when R_{det} is close to zero; in that regime the local Taylor expansion need not describe the heavy-tailed reciprocal statistic.

6 What conditioning on the design omits

Conditioning on X is an exact first step. The incomplete step is replacing all X -dependent conditional means by deterministic limits while retaining only response-noise variance. The law of total variance is

$$\text{Var}_{X,\epsilon}(R) = \mathbb{E}_X[\text{Var}_{\epsilon}(R | X)] + \text{Var}_X(\mathbb{E}_{\epsilon}[R | X]). \quad (52)$$

The current correction targets primarily the first term. The second term is a random-design fluctuation of the ridge resolvent.

This distinction is invisible to a leading deterministic equivalent: both terms can be $O(n^{-1})$ and therefore vanish as $n \rightarrow \infty$, while their sum is precisely the leading quantity needed for a second-order correction. A sharp diagnostic is $\sigma = 0$. Equation (45) then returns zero, although finite-sample variation of $P_X \beta^*$ across training sets generally gives nonzero $\text{Var}_X(R)$.

The covariance substitution in Eq. (36) is also simplified. Deterministic equivalents for derivatives of the resolvent introduce the susceptibility

$$\kappa'(\lambda) = \frac{1}{1 - \text{df}_2/n}. \quad (53)$$

A minimal refinement should carry such factors in noise bilinear forms. However, simply multiplying every term by κ' is not a complete solution: the quadratic trace and design covariance require joint two-resolvent equivalents.

7 Formulation of the full unconditional variance

Let

$$\mathbf{Z} = \begin{pmatrix} N \\ D \end{pmatrix}, \quad \bar{\mathbf{Z}} = \begin{pmatrix} \bar{N} \\ \bar{D} \end{pmatrix} = \mathbb{E}_{X,\epsilon} \mathbf{Z}. \quad (54)$$

To leading second order,

$$\text{Var}(R) \simeq \mathbf{J}^\top \mathcal{C}_{ND} \mathbf{J}, \quad \mathbf{J} = \begin{pmatrix} \overline{D}^{-1} \\ -\overline{ND}^{-2} \end{pmatrix}, \quad (55)$$

where

$$\mathcal{C}_{ND} = \text{Cov}_{X,\epsilon} \begin{pmatrix} N \\ D \end{pmatrix}. \quad (56)$$

The exact covariance decomposition is

$$\boxed{\mathcal{C}_{ND} = \mathbb{E}_X[\mathcal{C}_\epsilon(X)] + \mathcal{C}_X}, \quad (57)$$

with

$$\mathcal{C}_\epsilon(X) = \begin{pmatrix} (\boldsymbol{\beta}^\star)^\top \Omega_X \boldsymbol{\beta}^\star & 2(\boldsymbol{\beta}^\star)^\top \Omega_X \boldsymbol{\mu}_X \\ 2(\boldsymbol{\beta}^\star)^\top \Omega_X \boldsymbol{\mu}_X & 4\boldsymbol{\mu}_X^\top \Omega_X \boldsymbol{\mu}_X + 2\text{tr}(\Omega_X^2) \end{pmatrix}, \quad (58)$$

and

$$\mathcal{C}_X = \text{Cov}_X \begin{pmatrix} n_X \\ d_X \end{pmatrix}. \quad (59)$$

The current formula approximates selected entries of $\mathbb{E}_X[\mathcal{C}_\epsilon(X)]$ and sets $\mathcal{C}_X$ to zero.

7.1 Random-matrix objects required for the design covariance

Using $P_X = \widehat{\Sigma}(\widehat{\Sigma} + \lambda I)^{-1}$,

$$n_X = (\boldsymbol{\beta}^\star)^\top P_X \boldsymbol{\beta}^\star, \quad (60)$$

$$d_X = (\boldsymbol{\beta}^\star)^\top P_X^2 \boldsymbol{\beta}^\star + \frac{\sigma^2}{n} \text{tr}(Q_X \widehat{\Sigma} Q_X). \quad (61)$$

Since

$$P_X = I - \lambda Q_X, \quad P_X^2 = I - 2\lambda Q_X + \lambda^2 Q_X^2, \quad (62)$$

the design covariance in Eq. (59) requires a joint central-limit theorem for bilinear and trace resolvent statistics, including

$$(\boldsymbol{\beta}^\star)^\top Q_X \boldsymbol{\beta}^\star, \quad (\boldsymbol{\beta}^\star)^\top Q_X^2 \boldsymbol{\beta}^\star, \quad (63)$$

$$\text{tr}(Q_X \widehat{\Sigma} Q_X), \quad (64)$$

$$\text{Cov} \left((\boldsymbol{\beta}^\star)^\top Q_X \boldsymbol{\beta}^\star, (\boldsymbol{\beta}^\star)^\top Q_X^2 \boldsymbol{\beta}^\star \right), \quad (65)$$

$$\text{Cov} \left((\boldsymbol{\beta}^\star)^\top Q_X \boldsymbol{\beta}^\star, \text{tr}(Q_X \widehat{\Sigma} Q_X) \right), \quad (66)$$

and the remaining entries generated by Eq. (61). Equivalently, these can be obtained from a two-spectral-parameter covariance kernel for

$$(\boldsymbol{\beta}^\star)^\top (\widehat{\Sigma} - zI)^{-1} \boldsymbol{\beta}^\star$$

and its derivatives before taking $z = -\lambda$.

A complete calculation must specify:

1. deterministic means of N and D , including $O(n^{-1})$ bias if the mean of R is required to the same order;

2. the full 2×2 covariance matrix $\mathcal{C}_{ND}$ to order n^{-1} ;
3. fourth-cumulant terms for non-Gaussian designs;
4. spike or teacher-alignment terms when $\beta^\star$ is not generic with respect to the covariance eigenbasis.

For Gaussian designs the fourth-cumulant correction vanishes, but bilinear resolvent fluctuations remain.

7.2 Second-order correction to the mean ratio

If $\mathbb{E}R$, rather than only $\text{Var } R$, is needed to order n^{-1} , the second-order bivariate delta method gives

$$\mathbb{E}R \simeq \frac{\bar{N}}{\bar{D}} - \frac{\text{Cov}(N, D)}{\bar{D}^2} + \frac{\bar{N} \text{Var}(D)}{\bar{D}^3}. \quad (67)$$

Thus a full correction to $\mathbb{E}[(1 - R)^2]$ should use both the corrected mean in Eq. (67) and the full variance in Eq. (55).

8 Additional randomness from cross-validation

The fixed- λ theory above treats the regularization as deterministic. For RidgeCV, the fitted model is

$$\hat{\beta} = \hat{\beta}_{\hat{\alpha}(X, \epsilon)}, \quad (68)$$

so the selected grid point is another random variable. The unconditional variance contains:

1. fluctuations of N and D at fixed α ;
2. variation of $\hat{\alpha}$ across data and response noise;
3. covariance between $\hat{\alpha}$ and the downstream ratio.

If the CV risk has a unique smooth interior minimum, a local approximation is

$$\hat{\alpha} - \alpha_\star \simeq -\frac{\hat{L}'_{\text{CV}}(\alpha_\star)}{L''_{\text{CV}}(\alpha_\star)}. \quad (69)$$

One could augment the joint delta calculation with $(N, D, \hat{\alpha})$. For a discrete grid or a flat risk curve, the local approximation in Eq. (69) is unreliable; the appropriate description is a mixture over selected grid points,

$$\text{Var}(R) = \sum_a p_a \text{Var}(R \mid \hat{\alpha} = a) + \text{Var}_a(\mathbb{E}[R \mid \hat{\alpha} = a]). \quad (70)$$

9 Linear-feature generalization

For a linear feature map, let the feature coefficient be $\hat{\theta} = \mu + \xi$, with conditional covariance Ω . Pixel back-propagation introduces

$$G = AA^\top, \quad \mathbf{q} = A\beta^\star, \quad (71)$$

and the own-path ratio is

$$R = \frac{\widehat{\boldsymbol{\theta}}^\top \mathbf{q}}{\widehat{\boldsymbol{\theta}}^\top G \widehat{\boldsymbol{\theta}}}. \quad (72)$$

The fixed-design delta calculation becomes

$$\boxed{\text{Var}(R \mid X) \simeq \frac{\mathbf{v}^\top \Omega \mathbf{v} + 2R^2 \text{tr}[(G\Omega)^2]}{D^2}, \quad \mathbf{v} = \mathbf{q} - 2RG\boldsymbol{\mu}.} \quad (73)$$

The pixel-space formula is recovered with $G = I$ and $\mathbf{q} = \boldsymbol{\beta}^*$. For noncommuting feature covariance and back-propagation geometry, the full unconditional theory requires weighted two-resolvent covariances.

10 Practical validation and implementation roadmap

A nested simulation can measure every component of Eq. (52) without theoretical ambiguity:

1. draw B_X independent training matrices;
2. for each fixed X_b , draw B_ϵ independent response-noise vectors;
3. estimate the within- X variance and average it over datasets;
4. estimate the between- X variance of the conditional means;
5. repeat first at fixed α , then with RidgeCV.

This separates response noise, random design, and selection variability.

A staged theoretical implementation should proceed as follows:

1. replace the simplified noise covariance in Eq. (36) with its derivative-resolvent DE;
2. derive or import the joint Gaussian-design CLT for the bilinear forms in Section 7;
3. assemble $\mathcal{C}_{ND}$ using Eq. (57);
4. apply Eqs. (55) and (67);
5. validate fixed- α predictions with nested Monte Carlo;
6. add CV-selection variability only after the fixed- α variance is validated.

11 Summary

The current code formula is not an arbitrary spectral expression. It follows from four steps: conditional Gaussian ridge noise, a delta method for N/D , Wick identities, and a leading spectral DE substitution. It estimates response-noise fluctuations around a deterministic ridge fit. The full $\text{Var}(R)$ additionally needs the joint random-design covariance of the numerator and denominator resolvent statistics, second-order correction to the mean ratio, and—for RidgeCV—the distribution and covariance of the selected regularization.